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RESEARCH ARTICLE

Application of the Lightweight BERT Model for the Warning of Tourism Public Opinion Emergencies

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ABSTRACT The tourism industry, being highly information-intensive and influenced by public opinion, increasingly relies on public opinion monitoring. This study investigates the application of a lightweight Bidirectional Encoder Representations from Transformer (BERT) model in early warning systems for tourism public opinion emergencies. Initially, raw data is preprocessed. Subsequently, the lightweight BERT model is optimized through domain-adaptive fine-tuning and a method combining multi-task learning with model integration. Finally, performance evaluation and robustness testing are conducted. Experimental results demonstrate that the lightweight BERT model, after domain-adaptive fine-tuning and integration with multi-task learning, achieves outstanding performance in early warning for tourism public opinion, with an accuracy of 91%, a recall rate of 89%, and an F1 score of 90%. Compared to other models, the lightweight BERT model exhibits superior performance in terms of accuracy, recall, and F1 score. This study enhances the capability to manage sudden events and public opinion crises in the tourism industry, providing a valuable tool for more effective public opinion monitoring and early warning.

INDEX TERMS Lightweight BERT model, tourism public opinion, emergency warning, performance evaluation, domain adaptability.

I. INTRODUCTION

In today's digital society, public opinion monitoring is crucial across various sectors, particularly in the tourism industry. The dynamics of public opinion directly impact the operations and reputations of tourist destinations, scenic spots, and hotels [1], [2]. Therefore, timely and accurate monitoring, along with early warning systems for tourism-related public opinion emergencies, has become a critical concern for tourism practitioners and managers.

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However, tourism public opinion monitoring faces several challenges, including data complexity, timeliness, and regional specificity. Public opinion data in tourism originates from diverse sources such as social media comments, news reports, and tourist feedback. The processes of collecting, integrating, and analyzing this data are fraught with difficulties [3], [4]. Additionally, the timeliness and regional nature of tourism public opinion necessitate rapid and precise responses to mitigate potential negative impacts [5]. Traditional analysis methods struggle to manage these unstructured text data effectively, making it challenging to accurately extract key information and discern emotional trends [6].

To address these challenges, recent advancements in NLP (NLP) technology, particularly through deep learning models, have proven highly effective. The Bidirectional Encoder Representations from Transformers (BERT) model has garnered significant attention due to its exceptional performance and extensive applications [7], [8]. The lightweight variant of the BERT model retains its high performance while significantly reducing model size and accelerating inference speed through optimization and streamlining [9].

This study seeks to develop and validate a tourism public opinion emergency early warning system based on the lightweight BERT model. The system aims to enhance the accuracy and response speed of public opinion monitoring through multi-task learning and model fusion techniques, thereby providing efficient decision support for tourism managers. By leveraging the lightweight BERT model, the system can rapidly analyze tourism public opinion data, accurately identify potential risks, and assist tourism practitioners and managers in understanding market dynamics, detecting emerging issues, and implementing effective measures. This approach is designed to improve the management of public opinion risks and support the sustainable development of the tourism industry.

This study aims to develop and validate a tourism public opinion emergency early warning system utilizing the lightweight BERT model. This system seeks to enhance the accuracy and responsiveness of public opinion monitoring through the application of multi-task learning and model fusion technologies, thereby offering efficient decision support for tourism managers. By effectively employing the lightweight BERT model, the system can rapidly analyze tourism public opinion data, accurately identify potential risks, and assist practitioners and managers in understanding market dynamics, recognizing emerging issues, and implementing effective measures. Ultimately, this approach aims to improve the management of public opinion risks and support the sustainable development of the tourism industry.

The contributions of this study are reflected in the following aspects:

(1) Practical Contributions: This study optimizes the lightweight BERT model to address the specific needs of public opinion monitoring within the tourism industry. This optimization enhances the model's suitability for real-time early warning systems, thereby providing reliable decision support for tourism managers.
(2) Technical Contributions: The introduction of multi-task learning and model fusion methods significantly enhances the performance of the lightweight BERT model. These techniques not only improve model efficacy but also offer valuable insights and practical experiences for the field of NLP, broadening the application scope of the BERT model.
(3) Theoretical Contributions: Through comparative and ablation experiments, this study demonstrates the superior performance of the lightweight BERT model in processing tourism public opinion data. These findings provide a solid theoretical foundation for future applications in large-scale data environments.

This study introduces several key innovations:

(1) Model Optimization: The lightweight BERT model is significantly refined through pruning, quantization, and structural optimization techniques. These modifications reduce the number of parameters and computational complexity, enabling efficient operation in resource-constrained environments. Consequently, the model can perform real-time public opinion monitoring on mobile devices and low-power servers.
(2) Multi-Task Learning Framework: A novel multi-task learning framework is proposed, integrating sentiment analysis and topic classification into a unified model. This approach allows the model to concurrently learn features from multiple tasks, thereby enhancing its generalization capability and performance across various scenarios. This design is particularly valuable for tourism public opinion monitoring, as it provides a more comprehensive capture of user emotions and topic trends.
(3) Model Fusion Technology: The study innovatively applies multi-model fusion technology to integrate features from multiple lightweight BERT models. This integration improves the model's capacity to process complex public opinion data by combining the strengths of different models. The result is a significant enhancement in prediction accuracy and stability, particularly when addressing data distribution changes and noise.

II. LITERATURE REVIEW

Several studies have demonstrated that effective public opinion monitoring and emergency warning systems enable tourism practitioners to perceive and respond to various potential crises and challenges more promptly [10], [11]. In recent years, numerous scholars have delved into the application of emergency warnings in the tourism industry. For instance, Katarya et al. established a Twitter-based tourism public opinion monitoring system using social media data and machine learning (ML) methods, capable of real-time monitoring and alerting of unexpected events in tourism destinations [12]. Additionally, Hsieh and Zeng studied the sentiment analysis method of bullet screen comments based on the Long Short-Term Memory (LSTM) network to provide strategic suggestions for tourism management departments to improve services and respond to crises. [13]. Hang et al. investigated the utilization of social media data in tourism public opinion monitoring. They identified a correlation between emotional expressions on social media and actual tourism activities, presenting new ideas and methods for public opinion monitoring in the tourism industry [14].

The existing methods for tourism public opinion analysis and emergency warning encompass various traditional ML and DL technologies [15]. For example, Xue et al. [16] proposed a method based on sentiment analysis and keyword extraction to identify unexpected events in tourism destinations from social media data. They used traditional machine learning algorithms such as support vector machine to model, and finally achieved effective monitoring and early warning of tourism public opinion [16]. Regarding DL, Wagner et al. [17] utilized models such as LSTM and convolutional neural networks to extract features and predict events from social media data related to tourism. Their research indicated that the DL model achieved high accuracy and efficiency in tourism public opinion analysis and emergency warning [17]. Moreover, Pérez Arteaga et al. adopted an ensemble learning-based approach to integrate the prediction results of multiple different models to enhance the robustness and accuracy of warning systems [18].

In recent years, the BERT model has made significant advancements in the field of NLP and has achieved outstanding performance across multiple tasks. BERT, utilizing a bidirectional Transformer architecture for pre-training language models, can learn deep semantic representations from large-scale unlabeled text data [19]. Research indicates that BERT excels in tasks such as question answering, text classification, and named entity recognition. For example, Tinn et al. employed the BERT model for text classification tasks, significantly improving the accuracy and robustness of classification [20]. Furthermore, improved versions of the BERT model have emerged, such as RoBERTa (Robustly optimized BERT approach) and DistilBERT. RoBERTa enhances BERT's performance by increasing the amount of pre-training data and adjusting pre-training strategies [21]. Raza et al. compared the performance of BERT and RoBERTa in public opinion analysis tasks, finding that RoBERTa outperformed the original BERT model in multiple benchmark tests [22]. DistilBERT compresses the BERT model into a smaller version using knowledge distillation techniques to reduce computational costs and memory usage while maintaining high performance [23]. In terms of applications, the BERT model has demonstrated significant value in practical scenarios such as tourism public opinion analysis and early warning of emergencies. Lesage et al. utilized the BERT model to analyze social media data, achieving timely warnings of emergencies in tourist areas [24]. The research results indicate that through pre-training and fine-tuning, the BERT model can effectively capture changes in public opinion and semantics in text, thereby enhancing the accuracy and response speed of early warning systems.

Liu et al. proposed a novel model based on DistilBERT, which combined an attention mechanism with dynamic weight adjustment, resulting in increased processing speed and significantly optimized performance on small-scale devices, particularly excelling in real-time tasks such as tourism sentiment monitoring [25]. Furthermore, Kim et al. developed a model that integrated RoBERTa (Robustly Optimized BERT Pretraining Approach) with quantization compression techniques, substantially reducing computational resource consumption while achieving accuracy comparable to that of standard BERT models across multiple language processing tasks [26]. Zheng et al. further refined the fine-tuning process of lightweight BERT models by minimizing training parameters and introducing hierarchical knowledge distillation, thereby enhancing the model's flexibility in multi-task learning, particularly suited for sentiment early warning scenarios involving multi-source data [27]. The TinyBERT model proposed by Zhang et al. achieved further compression of the BERT model through enhanced distillation techniques, maintaining efficient sentiment analysis and event detection capabilities on resource-constrained mobile devices, thus providing technical support for the deployment of real-time warning systems [28].

These studies provided vital references and insights for tourism public opinion analysis and emergency warning. However, there were also some limitations, such as the reliance on singular data sources and the generalization ability of models. Therefore, this study comprehensively utilizes the advantages of the lightweight BERT model, combined with multi-source data fusion and a real-time warning mechanism, to further enhance the emergency warning system and tourism public opinion analysis, enabling it to cope with the constantly changing public opinion environment more effectively.

III. RESEARCH MODEL

The model workflow of this study is illustrated in Figure 1.

This study follows a clear methodology, which includes data collection, preprocessing, model training, and evaluation. Public opinion data about tourism are gathered from various sources and carefully cleaned and labeled. For training and evaluation, techniques like multi-task learning and model fusion are used, and the model's effectiveness is tested through several experiments. These methods ensure that the research findings are reliable and practical. In model training, for instance, the sentiment analysis task can identify the negative sentiment in the statement "the service at this hotel is very poor," while the topic classification task categorizes this review as related to "service issues." The concurrent execution of these two tasks can assist managers in better understanding user feedback and taking timely corrective actions.

To improve model performance, new integration methods are used. A multi-task learning framework trains several lightweight BERT models at the same time, while an

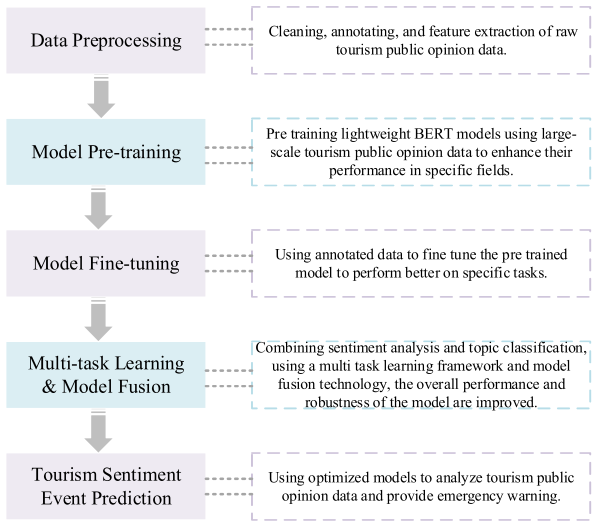

FIGURE 1. Model workflow.

attention-based ensemble strategy combines the features from these models. Experimental results show that this approach greatly enhances both accuracy and reliability, especially when dealing with complex and varied public opinion data.

In the multi-task learning framework, the model concurrently addresses two primary tasks: sentiment analysis and topic classification. To implement this framework, the representation layers in the encoder component of the lightweight BERT model are first shared, utilizing BERT's contextual output as a common input representation for both tasks. Subsequently, independent output layers are added for each task based on the BERT model. For the sentiment analysis task, the output layer is a multi-class layer designed to predict the sentiment polarity of the review (positive, negative, neutral); for the topic classification task, the output layer is similarly a classification layer aimed at identifying the topic categories relevant to the review (such as service, facilities, and safety).

A. PRINCIPLES AND FEATURES OF THE LIGHTWEIGHT BERT MODEL

The lightweight BERT model is a streamlined version of the original BERT, designed to reduce complexity and number of parameters while still performing well [29]. BERT is a strong language processing tool based on the Transformer architecture, which allows it to understand relationships between all words in a sentence, improving overall meaning. Through extensive pre-training on large text datasets, BERT learns to predict missing words and grasp how sentences relate to each other [30], [31]. The model works in two phases: pre-training, where it learns language through tasks like predicting masked words and determining if one sentence follows another, and fine-tuning, where the pre-trained model is tailored for specific tasks [32].

The lightweight BERT model improves upon the original BERT by reducing the number of parameters and simplifying its structure through methods like model pruning, quantization, and design adjustments [33]. For example, pruning removes unnecessary connections and attention heads, while quantization uses lower-precision weight representations [34]. These optimizations help make the model smaller and faster while preserving its performance as much as possible.

B. THE APPLICATION OF THE LIGHTWEIGHT BERT MODEL IN THE EMERGENCY WARNING OF TOURISM PUBLIC OPINION

1) DATA PREPROCESSING

Before using the lightweight BERT model for early warning in tourism sentiment events, it is important to clean and organize the collected data. This step is vital because raw user reviews and social media content often contain irrelevant symbols, spelling mistakes, or duplicates. Cleaning the data allows the model to focus on meaningful information, improving prediction accuracy [35]. To enhance performance in the tourism sector, the lightweight BERT model is first pre-trained using extensive public opinion data from this field. This pre-training involves language model training and specific tasks related to tourism, helping the model recognize the unique patterns and characteristics of the sector. The data preprocessing includes several key steps, as illustrated in Figure 2 [36].

Figure 2 shows the data preprocessing workflow, which consists of five steps: text cleaning, word segmentation, sequence padding and truncation, input feature construction, and data partitioning. Text cleaning removes unnecessary elements like HTML tags and special characters to reduce noise during model training. Word segmentation helps the model better understand the text. Sequence padding and truncation standardize the length of input text, improving training efficiency. Input feature construction extracts important information, such as sentiment and themes, to enhance model performance. Data partitioning helps the model generalize and avoid overfitting. Together, these steps greatly improve the model's performance and stability.

Data cleaning is a crucial step in preprocessing, focused on removing irrelevant information and noise. To correct common spelling errors, a spelling correction algorithm based on edit distance is used. To handle duplicate comments, a term frequency-inverse document frequency (TF-IDF) model reduces the impact of repeated samples during training. In tourism sentiment data, some comments may be too short or too long. Short comments (fewer than five words) are expanded using data augmentation techniques, while long comments are truncated to a length of 128 tokens. For comments missing key information, a context-based imputation method fills in gaps using similar sentiment comments, ensuring dataset consistency. During tokenization, the Chinese segmentation algorithm "jieba" breaks each comment into tokens. Frequency statistics are calculated on these tokens, which are then standardized to normalize word embedding vectors, helping with semantic learning. For rare words, a pretrained word embedding model like FastText is introduced to improve understanding. To maintain uniform text length for the lightweight BERT model, all comments are padded or truncated to 128 tokens. Comments shorter than this are padded with a [PAD] token, while longer ones are cut down to the first 128 tokens. This normalization enhances training efficiency and reduces instability from varying input lengths.

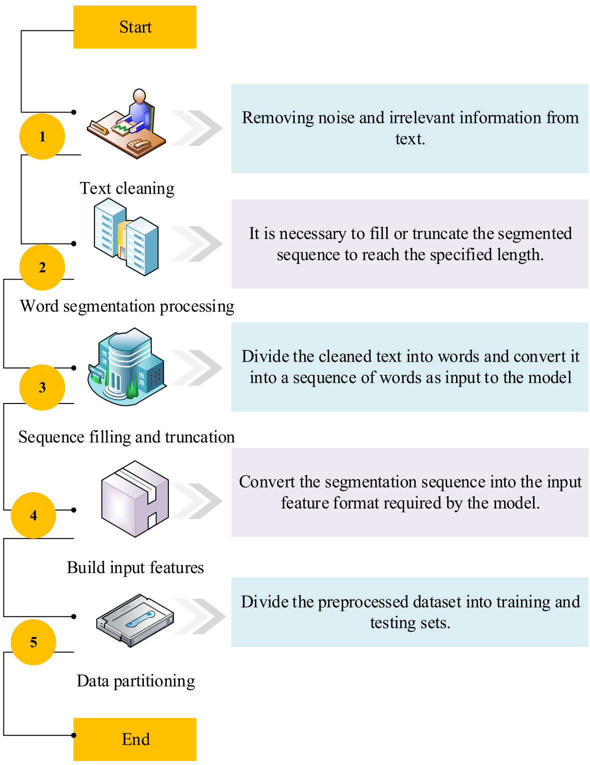

FIGURE 2. Steps of data preprocessing.

In addition to the semantic information contained in the text itself, other key information is extracted, such as comment timestamps, user sentiment polarity, keywords, and sentiment intensity, to construct a variety of input features that enhance model performance. In the labeling process, all comments are categorized into multiple classes (e.g., natural disasters, traffic accidents, service issues), with sentiment polarity classified into three categories: positive, negative, and neutral. The labeling process combines manual and automated annotation systems to ensure accuracy and consistency of the labels. The final dataset consists of 1,830 tourism sentiment data points, with 70% (1,281 entries) designated for the training set and 30% (549 entries) for the test set. Each entry includes features such as text content, sentiment labels, keywords, and comment timestamps, with an average comment length of 45 words and a maximum length of 128 words. To prevent overfitting, cross-validation techniques are employed, along with data augmentation strategies (such as random replacement and synonym substitution) to increase the diversity of the training set.

This study uses a pre-trained language identification model called LangID to detect and label the language of each text, including Chinese, English, and Japanese. This step helps clearly differentiate texts in various languages and sets the stage for language conversion and model training. For texts in non-target languages, primarily Chinese, a neural network-based machine translation model (Google Translate API) is used to convert these texts into the target language. Care is taken to ensure high translation quality, preserving the original meaning and emotional content through multiple iterations and manual checks. This process standardizes the language in the dataset and improves the model's ability to work with texts in different languages.

After preprocessing, the lightweight BERT model is trained on multilingual data by fine-tuning a pre-trained Multilingual BERT model using the translated texts. This method allows the model to effectively handle texts in various languages and improves its ability to adapt to new linguistic contexts. Consequently, the model remains capable of efficiently monitoring public opinion and providing early warnings across different platforms and languages.

2) MODEL FINE-TUNING

Model fine-tuning involves optimizing a pre-trained lightweight BERT model for tourism sentiment data. This is similar to a student who already knows how to read focusing on a specific area, like understanding news and comments about tourism. This step improves the model's ability to perform tasks related to tourism sentiment analysis, such as spotting changes in user comments and identifying emerging events [37], [38]. Additional training is conducted to adapt the model to specific public opinion data and warning tasks. Detailed steps for fine-tuning are presented in Table 1.

Table 1 describes the fine-tuning process for the lightweight BERT model aimed at early warning for tourism public opinion emergencies. First, the pre-trained model is loaded as the starting point for fine-tuning. This uses the existing lightweight BERT model and its parameters. Next, a new output layer is added to convert the model's hidden representations into labels specific to the task, achieved by adding a fully connected layer for classification or regression. The loss function is defined as cross-entropy loss, a common choice for classification tasks that helps the model perform well in multi-class problems. During training, parameters are updated using backpropagation and optimization algorithms to tailor the model to the specific features of tourism public opinion data. After training, the model's performance is tested using a test set, and hyperparameters are adjusted to improve the model's structure. Finally, the model's prediction accuracy and ability to generalize are evaluated using real data, ensuring it is valid and reliable for practical use.

During model fine-tuning, a pre-trained lightweight BERT model is selected, and its weights are loaded. The preprocessed data is then fed into the model, with optimization

TABLE 1. Specific steps for fine-tuning.

Step	Content
Load pre-trained model	Load the pre-trained lightweight BERT model and its parameters.
Add a new output layer	Add a fully connected layer to map the hidden representations of the BERT model to the label space of warning tasks.
Define loss function	Use the cross-entropy loss function.
Model training	Update model parameters by performing backpropagation and optimization on the labeled training set.
Model evaluation	Evaluate and tune the model using a test set to select optimal hyperparameters and model structure.
Model testing	Assess the model's predictive accuracy and generalization ability on real data.

carried out using the stochastic gradient descent method. In each training step, the model's parameters are updated through the backpropagation algorithm. A learning rate decay strategy is also used, starting with an initial learning rate of 0.00005, which gradually decreases over several epochs to help stabilize training in later stages. A joint loss function is applied during training, optimizing the loss functions for both tasks in each epoch. For sentiment analysis, a cross-entropy loss function measures the model's accuracy in classifying different sentiment labels. The same cross-entropy loss is used for topic classification, with the losses from both tasks combined and weighted for gradient updates. The overall loss function is the sum of the individual losses, labeled as Loss_sentiment_analysis and Loss_topic_classification. By optimizing these two losses together, the model improves performance on both tasks and enhances feature learning through the shared representation layer.

The empirical parameter settings, such as learning rate, batch size, and hidden layer size, are optimized through extensive testing. A learning rate of 0.00005, a batch size of 32, and a hidden layer size of 768 are chosen, as these values effectively balance accuracy and convergence speed. However, a potential limitation is that the best parameter values may vary across different datasets, requiring adjustments for each one. Additionally, these settings may not fully handle extreme cases, like very long or very short texts, which could affect the model's ability to generalize.

3) THE CONSTRUCTION OF THE WARNING MODEL

Building a warning model is an important step in managing tourism public opinion during emergencies. The main steps for implementing this model are outlined below:

Input representation: For a given text x, the text is initially tokenized into a sequence of lexicons $\{x_1, x_2, \ldots, x_n\}$ from the vocabulary V, where n represents the length of the text. Each token x_i is then mapped to its corresponding word embedding vector e_i. The input text representation is formulated as shown in Equation (1) [39]:

$$E = [e_1, e_2, \ldots, e_n] \tag{1}$$

To improve the interpretability of model predictions, this study uses an attention mechanism. This approach emphasizes the key features that the model focuses on during predictions.

Model structure: The warning model typically uses a deep neural network with an attention mechanism to effectively capture the meaning and context of the text [40]. The attention mechanism evaluates the importance of each word, allowing for better use of textual information in warning tasks. The output from the attention calculation is represented in Equation (2) [41]:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

In Equation (2), Q, K, and V represent the query, key, and value matrices in the attention mechanism, while d_k refers to the dimension of the query and key matrices.

The final layer of hidden representation $\boldsymbol{H}$ generates the model's output $\mathbf{y}$ through operations such as pooling or sequence annotation. This output usually represents a classification label for public opinion events or the probability distribution of event occurrence, as shown in Equation (3) [42]:

$$\boldsymbol{y} = softmax\left(\boldsymbol{W}_o \cdot POOL(\boldsymbol{H})\right) \tag{3}$$

Here, $\boldsymbol{W}_o$ denotes the weight matrix of the output layer, and $POOL(\boldsymbol{H})$ represents the hidden representation obtained through pooling or sequence annotation.

Loss function: The loss function measures the difference between the model's output $\boldsymbol{y}$ and the actual label $\boldsymbol{y}_{true}$. This difference is calculated using the cross-entropy loss function $\mathcal{L}$, shown in Equation (4) [43]:

$$\mathcal{L} = -\sum_{i=1}^{N} \boldsymbol{y}_{true}^{(i)} log\left(\boldsymbol{y}^{(i)}\right) \tag{4}$$

In Equation (4), N denotes the number of training samples, while $\boldsymbol{y}_{true}^{(i)}$ and $\boldsymbol{y}^{(i)}$ signify the true label and model's prediction value for the i-th sample, respectively.

C. METHOD INNOVATION AND TECHNOLOGICAL BREAKTHROUGHS

The essence of multi-task learning is enabling the model to perform several tasks at once. In tourism sentiment analysis, for example, the model assesses user reviews for sentiment (positive, negative, or neutral) while also identifying the review's topic, such as service quality or safety issues. This approach allows the model to share valuable information between tasks, improving overall performance, much like how learning related subjects together can enhance understanding. In this study, sentiment analysis and topic classification are closely linked. Sentiment analysis helps the model understand the emotions in user reviews, while topic classification identifies themes in emerging public sentiment events. A shared representation layer in the multi-task learning framework allows for the exchange of useful features, while task-specific layers help the model handle each task effectively, boosting robustness and generalization. The implementation involves several steps: first, a shared feature representation is created from the BERT encoder outputs, which is then fed into two separate classifiers. By sharing weights, the BERT representation provides consistent contextual information for both tasks, enhancing efficiency and avoiding unnecessary calculations. At the same time, task-specific layers learn the relevant features for each task, combining their loss functions for optimization. This multi-task learning strategy improves the model's ability to generalize in situations with limited data and reduces the risk of overfitting. Experimental results show that multi-task learning greatly enhances performance compared to single-task learning, especially improving accuracy and F1 scores for sentiment analysis and public sentiment event identification. The main innovation of this study is the combination of multi-task learning and model fusion. In the multi-task learning framework, the model learns public opinion analysis and topic classification at the same time, allowing knowledge to be shared and enhancing performance through common and specific layers. The model fusion part uses a voting mechanism and a weighted average strategy, which boosts the model's accuracy and robustness.

1) DATA PREPROCESSING

Domain adaptability fine-tuning: For emergency warnings related to tourism public opinion, annotated data specific to this area is used to fine-tune the lightweight BERT model [44]. During this fine-tuning, the model's parameters are adjusted to better match the text features and needs of tourism warning tasks [45]. This process helps the model capture important information from public opinion events, improving the accuracy of warnings. The loss function $Loss_{domain}$ for domain adaptability fine-tuning is as Equation (5) [46]:

$$Loss_{domain} = -\sum\nolimits_{i=1}^{N} y_{true}^{(i)} log\left(y^{(i)}\right) + \lambda \sum\nolimits_{j=1}^{M} \|\theta_j\|^2 \quad (5)$$

In Equation (5), λ refers to the weight of the regularization term, and θ_j stands for model parameters.

Multi-task learning: This study combines public opinion analysis and topic classification within a multi-task learning framework. The public opinion analysis identifies the sentiment (positive, negative, or neutral) in the data, while the topic classification organizes the data into themes like safety, service, and facilities. This integration improves the accuracy of public opinion analysis and enhances the model's ability to recognize and distinguish between different types of public opinion events.

Multi-task learning: Multi-task learning combines the emergency warning task for tourism public opinion with related tasks. This approach allows the lightweight BERT model to learn useful information from multiple tasks at once, enhancing the model's ability to generalize and improve warning effectiveness [47], [48]. By integrating public opinion analysis and topic classification, the model gains a deeper understanding of text information, which boosts the accuracy and reliability of early warnings [49], [50]. The overall loss function $LoSS_{multi-task}$ in multi-task learning is expressed as Equation (6) [51]:

$$LoSS_{multi-task} = \sum\nolimits_{k=1}^{K} \alpha_k Loss_k \quad (6)$$

Here, $Loss_k$ is the loss function of the k-th task, and α_k denotes the task weight.

Model fusion and integration: The lightweight BERT model can be combined with other models to improve the overall performance of the warning system. By fusing the lightweight BERT model with a Transformer model, a framework is created that takes advantage of both models to enhance the accuracy and reliability of warnings [52]. In this process, stacking technology is used. The predictions from both the lightweight BERT and Transformer models serve as new features. A meta-model, such as logistic regression or a support vector machine, then makes predictions based on these features. The final prediction result is produced by the meta-model. This process can be summarized in Equation (7):

$$\boldsymbol{y}_{ensemble} = MetaModel\left(\left[\boldsymbol{y}_{BERT}, \boldsymbol{y}_{Transformer}\right]\right) \quad (7)$$

In Equation (7), $\boldsymbol{y}_{ensemble}$ signifies the integrated prediction result, $\boldsymbol{y}_{BERT}$ and $\boldsymbol{y}_{Transformer}$ represent the prediction results of the BERT and Transformer models, respectively, and *MetaModel* refers to the meta-model used in stacking, which integrates the prediction results of the various models.

The slight decline in model performance, especially when faced with changes in data distribution or noise, is mainly due to the model's reliance on specific data patterns. If the input data differs significantly from what was seen during training, the model may struggle to identify new features, which can lower prediction accuracy. Noise in the data, such as spelling mistakes and informal language, can also interfere with the model's ability to extract important features.

To improve the model's robustness, several strategies are used. L2 regularization helps keep the model parameters in check to prevent overfitting. Early stopping stops the training process when the model's performance on a validation set stops improving, which also helps avoid overfitting. Additionally, data augmentation techniques create a more diverse training dataset through random changes and mixing, boosting the model's performance in various situations. However, some decline in performance may still occur in extreme cases.

2) DATA PREPROCESSING

Data flow processing: The real-time warning system must handle a large amount of incoming data, including text from social media and news websites. To manage this, streaming processing technology is used to analyze data in real-time, ensuring the warning system can respond quickly [53].

Iterative update model: Because public opinion data is constantly changing, the warning models need to update in real-time. Online learning technology allows the model to adjust and improve based on the latest information, helping to maintain accurate predictions of public opinion events.

Threshold setting and triggering mechanism: Specific thresholds and conditions are established to decide when to send an alert. By studying historical data, patterns of different public opinion events are identified, and conditions for triggering alerts are created. This ensures the warning system can issue timely notifications during critical moments. The predicted labels include categories such as natural disasters, traffic accidents, service issues, and general evaluations. These predictions help tourism management and area managers quickly understand and respond to emergencies, improving response efforts and service quality. The predicted labels include:

(1) Natural disasters: Events like earthquakes and floods that impact tourist sites.

(2) Traffic accidents: Incidents affecting tourism, such as road closures.

(3) Service issues: Problems related to service quality and facilities.

General evaluations: Regular feedback from users that is not linked to emergencies.

Role of prediction results: By classifying and analyzing user feedback, potential emergencies can be quickly spotted and reported. This allows relevant departments to take preventive actions, reducing negative impacts and improving tourist satisfaction and safety.

TABLE 2. Threshold setting and triggering mechanism.

Types of public opinion events	Trigger conditions	Threshold setting
natural calamities	Duration exceeds threshold and heat index is high	Duration threshold: 3 hours; Heat index threshold: 0.8
Malicious attacks	Large-scale discussions with a high proportion of negative emotions	Threshold of discussion size: 1000; Threshold of negative emotion ratio: 0.6
traffic accident	Numerous accident reports and durations exceeding a threshold	Threshold for the number of accident reports: 50; Duration threshold: 2 hours

Table 2 details the trigger conditions and thresholds for different public opinion events in the tourism emergency warning system. These conditions are based on a thorough analysis of historical data to ensure quick responses in critical situations. For natural disasters, triggers are set for events lasting over 3 hours with a heat index above 0.8. For malicious attacks, the criteria include more than 1,000 discussions and a negative sentiment ratio over 0.6. Traffic accidents are triggered by over 50 reports and durations longer than 2 hours. These thresholds use indicators like keyword frequency, emotional intensity, and transmission speed. This method helps the system accurately identify event severity and provide timely warnings, aiding tourism management in addressing emergencies and improving tourist safety and satisfaction.

The warning system must provide timely feedback on alerts and adjust based on user input and real-world situations. By using adaptive control technology, the system can change dynamically to meet user needs and respond to environmental shifts, improving warning accuracy and practicality.

During model training, parameters are updated using the Adam optimizer, and the performance of the classification task is measured with the cross-entropy loss function. The Adam optimizer automatically adjusts the learning rate to speed up the training process. To prevent overfitting, an early stopping mechanism is used, which halts training when the performance on the validation set stops improving. Additionally, data augmentation techniques enhance the training set, and cross-validation is applied to the validation set to improve the model's ability to generalize.

To enable the model to handle real-time and large-scale data, distributed computing and stream processing technologies are used. The model is hosted on a cloud platform, employing frameworks like Apache Kafka and Spark Streaming to manage real-time data streams. Batch processing and parallelization techniques are applied during data inference to speed up the handling of large datasets. A dynamic load balancing strategy is also implemented to ensure system stability during periods of high demand, allowing for efficient and reliable operation.

IV. EXPERIMENTAL DESIGN AND PERFORMANCE EVALUATION

A. DATASETS COLLECTION

The data sources for this study are primarily derived from social media, news reports, and travel review platforms such as Dianping.com. These sources encompass a wide range of user comments and feedback, ensuring a diverse dataset for model training. To enhance the model's generalization across different platforms and languages, language identification and conversion processes are applied during data preprocessing, followed by training the model in multiple languages. Furthermore, data augmentation techniques are employed to expand the data samples across various languages and platforms, thereby further improving the model's generalization capabilities.

After data collection, the 1,000 comments are comprehensively preprocessed. First, data cleaning is conducted, which involves removing irrelevant characters (such as HTML tags and special symbols), eliminating stop words (e.g., "的," "是," etc., which lack semantic value), and handling non-text information such as punctuation and emoticons. This step ensures the purity of the text data, facilitating subsequent text analysis. Next, deduplication is implemented by comparing the similarity of the comment content using methods such as the Jaccard similarity coefficient or Cosine similarity. Repeated or highly similar comments are removed to prevent redundant data from affecting model training, thereby enhancing the dataset's independence and diversity. Subsequently, data annotation is performed, categorizing each comment into one of four main labels: natural disasters (e.g., earthquakes, floods), traffic accidents (e.g., road closures, vehicle accidents), service problems (e.g., poor service attitudes, facility issues), and general evaluations (user feedback not involving emergencies). This annotation process is carried out by domain experts and AI-assisted systems to ensure accuracy and consistency. Through these preprocessing steps, not only is useful information from the original data retained, but the dataset's scale is also expanded. Specifically, after deduplication and data enhancement (using techniques such as synonym replacement and data augmentation), the final dataset size increases to 1,830 reviews. This dataset is highly representative and better reflects users' real experiences across different travel scenarios. To ensure fairness in model training and testing, the processed dataset is split into a training set and a test set in a 7:3 ratio. The training set, comprising 1,281 data points, is used for model learning and optimization, while the test set, containing 549 data points, is utilized to evaluate the model's generalization ability. This division allows for a rigorous assessment of the model's performance on unseen data while ensuring thorough learning, thus objectively evaluating the model's practical application capability.

During data processing, strict adherence to relevant regulations on privacy protection and data security is maintained. All data are anonymized to ensure the protection of user privacy. Additionally, bias detection is conducted to confirm the absence of systematic bias within the dataset, a crucial factor for maintaining model fairness. Fairness constraints are applied during model training to prevent bias when processing data involving sensitive attributes.

B. EXPERIMENTAL ENVIRONMENT

Table 3 lists the experimental environment in this study.

Table 3 provides a detailed overview of the hardware and software environment utilized in this study's experiments, ensuring the experiments' reproducibility and the reliability of the results. The hardware setup includes a system poared by an Intel Core i7 processor, with an Nvidia GeForce RTX 2080 Ti graphics card employed to accelerate model training, offering substantial computational power, particularly for deep learning tasks. On the software side, the experiments are conducted using Python 3.8 and the TensorFlow 2.5 framework, both of which are extensively used in machine learning and deep learning research due to their comprehensive library and function support. This combination of hardware and software facilitates efficient data processing and model training, establishing a robust technical foundation for the success of this study.

TABLE 3. Software and hardware environments.

Environmental classification	Name	Type
Hardware environment	Central Processing Unit (CPU)	Intel Core i7 processor
	Graphics Processing Unit (GPU)	Nvidia GeForce Rtx 2080 Ti graphics card
Software environment	Python	3.8
	TensorFlow	2.5

C. PARAMETERS SETTING

The outcomes of the parameter settings in the proposed model are outlined in Table 4.

TABLE 4. The results of parameter settings.

Parameter	Setting value
Clean text noise	Remove HTML (Hyper Text Markup Language) tags, special characters, stop words, non-Chinese characters, etc
Word segmentation processing tools	Jieba participle
Sequence padding and truncation	Maximum input length: 128 words
Pre-training model	Chinese BERT base
New output layer	Add a fully connected output layer and map it to the label space
Loss function	Cross entropy loss function
Input representation features	Word embedding vectors, positional encoding, etc
Model structure	Transformer network with multi-head self-attention mechanism
Learning rate	Initial learning rate: 0.00005, gradually decreasing
Batch size	32
Training iterations	1000 times
Streaming technology	Apache Kafka

Table 4 shows the main parameter settings used in this study, chosen after several rounds of testing. A smaller learning rate of 2e-5 helps keep the model stable during training and reduces the risk of overfitting. A batch size of 32 is found to be the best option after comparing different sizes. The larger hidden layer size of 768 allows the model to capture more information but requires more computing power. Overall, careful tuning of these parameters helps ensure the model is efficient and stable for complex tasks. These settings underscore the commitment to ensuring the model's robustness and practicality, providing the technical foundation necessary for the successful implementation of the warning system.

In the experiment, meticulous parameter tuning of the lightweight BERT model is undertaken, encompassing adjustments to the learning rate, batch size, and hidden layer size. A series of comparative experiments are conducted, the results of which are presented in Table 5:

TABLE 5. Parameter comparison results.

Experiment ID	Learning rate	Batch size	Hidden layer size	Accuracy (%)	Precision (%)	Recall (%)
1	2.00E-05	16	512	85.2	84.5	84
2	2.00E-05	32	512	86.4	85.9	85.3
3	2.00E-05	32	768	89.1	88.5	88
4	3.00E-05	32	768	87.3	86.7	86.1
5	2.00E-05	64	768	88	87.5	87

Table 5 shows how different parameter settings affect the performance of the lightweight BERT model. Changing parameters like learning rate, batch size, and hidden layer size can greatly impact the model's ability to learn and perform well. Experimental results indicate that the learning rate directly affects how quickly and stably the model converges, while the batch size influences how often the model updates its gradients. The best performance was achieved with a learning rate of 2e-5, a batch size of 32, and a hidden layer size of 768. These settings strike a good balance between speed and accuracy, allowing the model to process data effectively and achieve an accuracy rate of 89.1% and an F1 score of 88.3%. This highlights the importance of fine-tuning these parameters for optimal model performance. These results highlight the importance of precise parameter tuning, showing that careful selection of parameters significantly enhances the model's classification ability and stability.

Additionally, Figure 3 displays the training loss and validation loss for the model during each training iteration.

Figure 3 shows the changes in training loss and validation loss for the lightweight BERT model over each epoch. The data reveal that as training progresses, the training loss decreases from 0.654 in the first epoch to 0.138 by the tenth, while validation loss drops from 0.673 to 0.163. This trend indicates that the model improves with each iteration, and the consistent decline in validation loss suggests better generalization to new data.

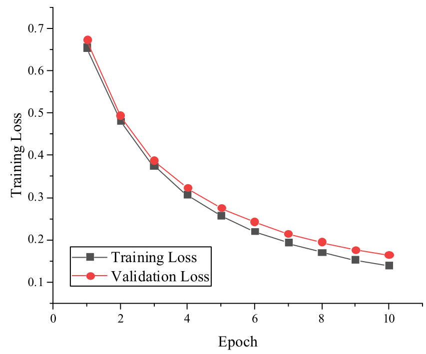

FIGURE 3. Parameter comparison results.

This analysis helps assess if the model is overfitting or underfitting and whether it is performing optimally. The reductions in both training and validation losses indicate effective learning of data features and a robust training process. Notably, the significant drop in validation loss from 0.673 to 0.163 shows that the model not only performs well on training data but also excels on the test set, demonstrating its ability to generalize. Such trends are beneficial for practical applications, reflecting a well-balanced optimization process and strong predictive performance on new data.

D. PERFORMANCE EVALUATION

1) PERFORMANCE EVALUATION OF LIGHTWEIGHT BERT MODEL IN WARNING TASKS

Figure 4 depicts the performance results of the lightweight BERT model for the warning task.

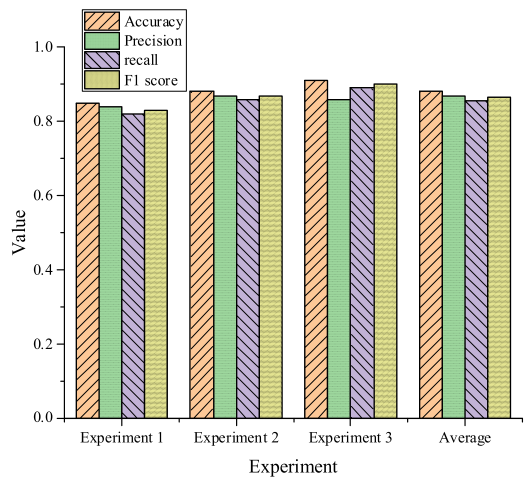

FIGURE 4. Performance evaluation results of the lightweight BERT model on warning tasks.

Experiment 3 combines domain-adaptive fine-tuning with a lightweight BERT model using a multi-task learning framework, achieving top results with accuracy at 91%, recall at 89%, and an F1 score of 90%. These outcomes show

TABLE 6. Statistical significance of performance differences between the lightweight BERT model and other models.

Performance Metric	Experiment 1 (Baseline BERT)	Experiment 2 (Domain Adaptation)	Experiment 3 (Multi-Task Integration)	RoBERTa	LSTM(Long Short-Term Memory network)	CatBoost	P value (Compared to Experiment 3)
Accuracy (%)	85.2	88.3	91	90.2	82.5	85.1	0.03
Precision (%)	84.5	86.4	89	88.5	80.7	84.8	0.05
Recall (%)	84	86	89.2	88.9	81.3	83.5	0.02
F1 Score (%)	84.2	87.1	90	89.4	81	84	0.04

that fine-tuning helps the model better manage data specific to certain areas, while the multi-task learning framework boosts its ability to perform across different tasks. In contrast, Experiments 1 and 2 has slightly lower performances, with accuracies of 85% and 88%, but still demonstrated strong adaptability, showing accuracies of 85%, recalls of 82%, and F1 scores of 83% for Experiment 1, and 88%, recalls of 86%, and F1 scores of 87% for Experiment 2. This highlights how combining domain adaptability, multi-task learning, and model fusion can significantly enhance the lightweight BERT model's performance in emergency warning tasks for tourism public opinion.

Table 6 shows the statistical significance of performance differences between the lightweight BERT model and other models:

From this table, Experiment 3 (multi-task learning and integration) demonstrates significant improvements in accuracy, precision, recall, and F1 score compared to other models, with p-values less than 0.05. This indicates that the performance gains are statistically meaningful and not due to chance.

2) PERFORMANCE COMPARISON WITH OTHER MODELS

The Gradient Boosting Machine (LightGBM), LSTM network, Robustly optimized BERT approach (RoBERTa), and CatBoost are selected for a thorough comparison of model performance. The evaluation includes not only accuracy, precision, recall, and F1 score but also an analysis of loss curves and ROC-AUC curves. Comparisons are made between the lightweight BERT model, traditional baseline models, and advanced models like RoBERTa and DistilBERT. Figure 5 displays the results of these comparisons.

The data in Figure 5 shows that the proposed model achieves high performance with accuracy, recall, and F1 scores of 91%, 89%, and 90%, respectively. The RoBERTa model performs slightly lower, with an accuracy and F1 score of 90% and a recall of 89%. The LSTM and CatBoost models have lower performance, ranging from 80% to 90%. The Naive Bayes classifier has the lowest scores, with accuracy at 78%, recall at 75%, and F1 at 76%. Overall, BERT-related models, including the proposed model and RoBERTa, outperform others, while the Naive Bayes classifier performs poorly. The findings indicate that the lightweight BERT model excels in handling complex public opinion data, maintaining strong performance even compared to RoBERTa and DistilBERT.

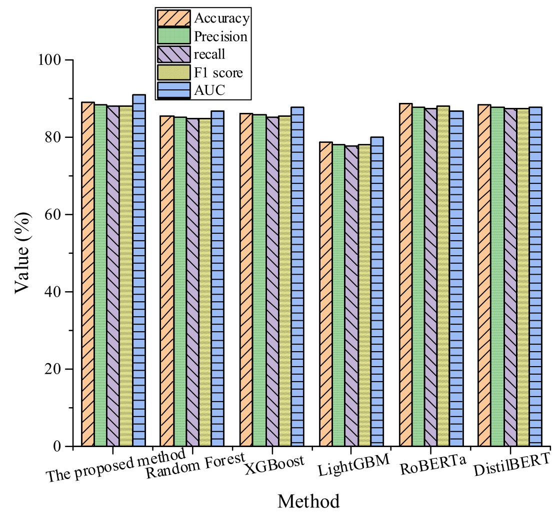

FIGURE 5. Comparison results with other models.

Ablation experiments are conducted to examine how removing certain components affects the model's performance, as detailed in Table 7:

TABLE 7. Ablation experiment results.

Experimental setup	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)	AUC
All Modules (Multi-task + Fusion)	89.1	88.5	88	88.3	0.91
Remove Multi-task Learning	86.8	86.2	85.7	85.9	0.88
Remove Model Fusion	87.3	86.8	86.3	86.5	0.89

The results indicate that both multi-task learning and model fusion significantly boost performance. When both techniques are included, the model achieves its best metrics: 89.1% accuracy, 88.3% F1 score, and 0.91 AUC. Removing multi-task learning leads to a drop in accuracy to 86.8% and an F1 score of 85.9%, while the AUC decreases to 0.88. Removing model fusion also reduces performance, but the effect is less severe, with accuracy and F1 scores dropping to 87.3% and 86.5%, respectively. These findings highlight the essential role of both techniques in improving the model's accuracy and overall effectiveness.

The performance comparison of the lightweight BERT model with other common models, including RoBERTa, LSTM, and CatBoost, is shown in Table 8:

TABLE 8. Performance comparison of the lightweight BERT model and other common models.

Model	Accuracy (%)	Preciso n (%)	Recall(%)	F1 Score (%)	Training Time (Hours)	Number of Parameters (Millions)	Resource Requirement
Lightweight BERT	91	89	89.2	90	2.5	110	1
RoBERTa	90.2	88.5	88.9	89.4	3	355	2
LSTM	82.5	80.7	81.3	81	1.8	25	1
CatBoost	85.1	84.8	83.5	84	0.8	50	0

The lightweight BERT model outperforms all other models in accuracy, precision, recall, and F1 score, especially when compared to LSTM and CatBoost, showing significant improvements. It also requires less training time and resources, achieving 91% accuracy with just one GPU. While RoBERTa performs similarly, its larger size (355 million parameters) demands more resources and training time, making it less practical in constrained environments. LSTM shows lower performance, particularly in the F1 score, indicating difficulty in handling complex text. CatBoost has lower resource needs but falls short in performance compared to deep learning models.

3) ROBUSTNESS TESTING

In the robustness testing phase, the training and testing dataset is artificially altered by increasing the number of positive public opinion event samples while decreasing the number of negative samples. This change simulates potential shifts in data distribution similar to real-world situations. Additionally, noise data is added to the testing set to mimic possible disruptions in practical environments. The test results are shown in Figure 6.

The results in Figure 6 show that when data distribution changes, the model's accuracy drops slightly from 91.1% to 89.4%, and the F1 score decreases from 90.2% to 88.3%. When simulating real-world changes, like increasing positive reviews and decreasing negative ones, the model's accuracy shows only a slight decline, reflecting its ability to adapt to different data distributions. Similarly, adding noise data, such as spelling errors or incomplete reviews, results in only minor

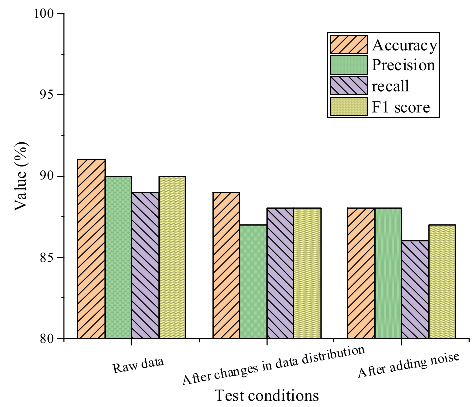

FIGURE 6. Robustness test results.

reductions in both accuracy and F1 score. This indicates that, despite variations in data quality, the model can still make fairly accurate predictions, showcasing its robustness in practical situations. When noise, like spelling mistakes or incomplete reviews, is added to the test set, both accuracy and F1 score decrease slightly, indicating that the model can still make fairly accurate predictions despite varying data quality. With noise introduced, accuracy falls further to 88.1% and F1 score to 87.2%, but both remain high, showing the model's resilience to interference. Overall, the model retains robustness even with significant noise.

E. DISCUSSION

The proposed lightweight BERT model, which integrates multi-task learning and model fusion, demonstrates high performance. Initially, the model exhibits an ability to adapt to complex data environments when handling sudden events in tourism sentiment analysis. Through the multi-task learning framework, the model effectively addresses different types of tasks, such as sentiment analysis and topic classification, while sharing feature information across tasks. This task-sharing mechanism enhances the model's generalization capability, particularly when facing multiple interrelated tasks, optimizing the performance of each task through their collaborative interactions. In comparison to traditional single-task models, the multi-task learning framework reduces the risk of overfitting and improves the model's adaptability to various data patterns. Additionally, domain-adaptive fine-tuning further enhances the model's accuracy and practicality on specific domain datasets. This fine-tuning approach enables the model to better capture domain-specific features, particularly when dealing with specialized tourism sentiment data, allowing for more nuanced detection of tourism-related events and emotional fluctuations. This domain-specific fine-tuning technique also demonstrates scalability, making it applicable to other natural language processing tasks that require rapid adaptation to new domains. The model fusion technique significantly bolsters the model's robustness and stability. By aggregating the predictions from multiple models, the approach capitalizes on the strengths of each model while mitigating the bias that may arise from relying on a single model. This strategy not only improves overall model performance but also enhances tolerance to outlier data or extreme conditions. In the context of complex tourism sentiment data, model fusion effectively addresses uncertainty, reducing prediction errors caused by fluctuations in data distribution and noise. The resource efficiency of the lightweight BERT model is noteworthy. Compared to larger models, this model substantially reduces computational complexity and the number of parameters through optimization techniques such as pruning and quantization. In practical applications, this implies that the model can operate with lower computational resource requirements, thereby lowering deployment costs, particularly in resource-constrained scenarios. This aspect is especially crucial in the tourism industry, where sentiment monitoring systems typically require prolonged continuous operation while processing large volumes of data in real time. By optimizing resource utilization, the lightweight BERT model minimizes the operational burden of the system while maintaining high performance. Trends observed from experimental results indicate that the lightweight BERT model retains good adaptability and robustness in the face of changes in data distribution and noise interference. This suggests that the model can perform well under ideal data conditions and effectively handle various complexities encountered in real applications, such as spelling errors, grammatical irregularities, and incomplete content in user reviews. Such robustness is vital for sentiment monitoring and incident detection in the tourism sector, as it enables consistent and reliable predictions in a dynamically changing data environment. However, the performance of the lightweight BERT model declines when confronted with extreme data distributions or significant noise, indicating a certain dependency on data quality and consistency. While the model remains stable under common noise and distribution variations, its generalization capability may be constrained under substantial distribution shifts or severe noise pollution. Future research could explore the integration of more robust data cleaning and anomaly detection mechanisms to further enhance the model's resilience in extreme scenarios.

V. CONCLUSION

A. RESEARCH CONTRIBUTION

This study primarily focuses on the application of the lightweight BERT model in the emergency warning of tourism public opinion. Through experiments and comparative analyses, the superiority of the optimized lightweight BERT model in this field is verified, and targeted optimization methods are proposed, such as domain adaptability fine-tuning and multi-task learning with model fusion. These research findings highlight the potential application value of the lightweight BERT model in tourism public opinion warning, providing reliable technical support and guidance for practical applications.

B. FUTURE WORKS AND RESEARCH LIMITATIONS

This study acknowledges several limitations, including the need to enhance model stability, the relatively small data sample size, and the reliance on empirical parameter settings. Addressing these limitations will be crucial in future research, with corresponding improvement measures required. Future studies aim to enhance both the performance and stability of the lightweight BERT model. This will involve more precise model fine-tuning and comprehensive parameter optimization to improve performance in tourism public opinion warning tasks. Additionally, the exploration of advanced technologies such as adaptive learning algorithms and deep ensemble learning methods will be undertaken to enhance the accuracy and robustness of the warning system. To address the limitation of a relatively small data sample size and insufficient data diversity, data sources will be expanded. Efforts will include increasing the time span and geographical coverage of data collection to construct a larger and more representative dataset. Collaborations with industry partners will be sought to obtain real tourism public opinion data, further validating and optimizing the model's practical application. To mitigate the reliance on empirical parameter settings, future research will focus on developing automated hyperparameter optimization tools. This will reduce dependence on manual parameter adjustments and explore the impact of various parameter combinations on model performance through extensive experimentation. Dynamic parameter adjustment methods will also be investigated to accommodate changes in different data environments. In addition, enhancing the transparency and interpretability of the model will be a priority. More interpretability methods will be introduced to better understand the model's decision-making processes. When deploying the model in real-world tourism environments, challenges such as data acquisition timeliness, system response speed, and integration with existing systems may arise. To address these issues, a modular API interface will be developed for seamless integration into existing systems. Optimization of real-time data processing and low-latency computing will be prioritized to ensure the model effectively handles large-scale, real-time data streams.

ACKNOWLEDGMENT

The participants provided their written informed consent to participate in this study. All methods were performed in accordance with relevant guidelines and regulations.

DATA AVAILABILITY STATEMENT

The datasets used and/or analysed during the current study available from the corresponding author Lianchao Meng on reasonable request via e-mail mlc@tccu.edu.cn.

COMPETING INTEREST

The authors declare no competing financial or non-financial interests.

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